# Notes: Belgibayeva, Bono, Bracke, Clara, Cocco & Majer (RFS, 2026)

## When Discounted Rates End: The Costs of Taking Action in the Mortgage Market

# Contents

# 1 Big picture

- **Question.** Why do many mortgage borrowers fail to remortgage when their discounted fixed-rate period ends, even though the monetary gains from action are large and relatively easy to compute?
- **Setting.** The paper studies the U.K. mortgage market, where many mortgages have an initial short discounted rate, commonly 2 years, and then roll onto a substantially higher reversion rate unless the borrower remortgages.
- **Empirical fact.** Among eligible 2-year fixed-rate borrowers originated between July 2013 and June 2014, about 19% remortgage externally, 47% remortgage internally, 20% remain on the reversion rate, and 14% repay the mortgage by December 2016.
- **Why the setting is useful.** Unlike U.S.-style long-term refinancing decisions, the end of a U.K. introductory period is salient, predetermined, and mechanically raises payments. This makes the gains from remortgaging unusually transparent.
- **Main mechanism.** Borrowers face nonpecuniary costs of action: remembering to act, comparing products, contacting lenders, switching lenders, and providing documentation for affordability checks.
- **Structural contribution.** The paper combines administrative data with a life-cycle mortgage model to infer the utility-equivalent costs needed to rationalize observed inaction.
- **Core result.** The inferred action costs are large. Measured relative to income, they are particularly high for younger and lower-income households.
- **Welfare result.** Eliminating action costs generates meaningful welfare gains. The largest gains come from reducing the broad action cost, not merely the cost of switching lender or passing affordability checks.
- **Takeaway.** Inaction in consumer finance is not explained only by calculation errors or small stakes. Even when monetary gains are salient, nonpecuniary action costs can prevent households from taking valuable actions.

# 2 Incremental contribution of the paper

- **Administrative measurement of all remortgaging outcomes.** The paper combines Product Sales Data, an FCA Request for Information, and lender loan-book data to observe external remortgages, internal remortgages, reversion, and repayment. This is unusually comprehensive for mortgage-market research.
- **Clean setting for inaction.** The U.K. discounted-rate reset creates a natural decision point with large, predictable monthly payment increases. This avoids the harder problem of estimating option values and interest-rate expectations in standard long-term refinancing models.
- **Separates internal and external remortgaging.** The paper distinguishes low-friction internal product transfers from higher-friction external remortgages that require lender switching and affordability checks.
- **Structural utility-cost measurement.** Instead of only documenting that borrowers leave money on the table, the paper estimates nonpecuniary costs in a life-cycle model with endogenous marginal utility of consumption.
- **Heterogeneity in action costs.** The paper shows that utility costs measured as income shares are larger for young and low-income borrowers, connecting mortgage inaction to financial stress and limited mental bandwidth.
- **Policy relevance.** The model quantifies welfare gains from reducing different frictions: action costs, switching costs, and affordability-check costs. This makes the paper useful for consumer-protection and mortgage-market policy.
- **Out-of-sample validation.** The paper applies the calibrated model to a later cohort facing different house-price and interest-rate conditions, providing an important check on the model's predictive power.

**Interpretation:** The incremental contribution is the combination of measurement and structural interpretation. The administrative data reveal the full set of borrower choices; the model explains how large utility costs must be to make those choices optimal.

# 3 Data and measurement

## 3.1 Institutional setting and remortgage options

A borrower whose discounted rate ends can take one of four broad paths:

1. **Inaction / reversion:** do not remortgage and move to the higher reversion rate.
2. **Internal remortgage without equity extraction:** switch to another product with the same lender, usually without affordability checks.
3. **Internal remortgage with equity extraction:** stay with the same lender but increase the loan balance, requiring action and affordability checks.
4. **External remortgage:** switch lenders, requiring action, affordability checks, and lender switching.

**Interpretation:** This hierarchy is important because different options require different nonpecuniary costs. Internal remortgages without equity extraction are the least demanding; external remortgages are the most demanding.

## 3.2 Administrative data sources

The paper uses three main administrative data sources:

- **PSD001:** origination data for regulated first-charge residential mortgages in the United Kingdom.
- **FCA RFI:** a special request to major lenders collecting internal remortgage information for 2015 and 2016.
- **PSD007:** semiannual loan-book data with balances, rate type, reversion status, arrears, and repayment information.

**Interpretation:** The unique value of the data is that it closes the measurement gap between external remortgages, internal remortgages, reversion, and repayment. Most mortgage datasets observe only some of these outcomes.

## 3.3 Sample construction

The main sample consists of borrowers who originated a 2-year fixed-rate mortgage between July 2013 and June 2014. These borrowers are followed through December 2016, which captures the period after the initial discounted rate expires. The authors exclude borrowers who could not reasonably remortgage because of small balances, short remaining maturities, arrears or forbearance, or very high LTV.

**Interpretation:** The exclusion restrictions are designed to focus the analysis on borrowers who had the ability to act. This makes observed inaction more informative about nonpecuniary costs rather than eligibility constraints.

### 3.4 Monetary gains from remortgaging

The paper estimates gains from remortgaging by comparing the cost of staying on the reversion rate to the cost of moving to an internal or external remortgage product, net of fees:

$$Gain = PV(\text{reversion payments}) - PV(\text{new mortgage payments}) - \text{fees}.$$

For internal remortgage options, the paper estimates the rate a borrower could obtain with the same lender using observed rates of similar borrowers. For external remortgages, it uses market offers available to comparable borrowers.

**Interpretation:** The estimated gains are high for many borrowers, which makes inaction puzzling under a model with low action costs.

### 3.5 Inaction and action-cost interpretation

Inaction is defined as remaining on the reversion rate by December 2016. The key regression and structural moments compare inaction to estimated monetary gains, income, age, and mortgage characteristics.

**Interpretation:** A reduced-form negative relation between gains and inaction shows that borrowers respond to incentives. But the high level of inaction at large gains shows that something beyond monetary gains matters. The structural model quantifies that something as nonpecuniary costs.

## 4 Empirical specifications and structural model

### 4.1 Reduced-form inaction regressions

The paper estimates regressions of the form:

$$Inaction_i = f(Gain_i, Income_i, Gain_i \times Income_i, X_i) + \varepsilon_i,$$

where $X_i$ includes origination characteristics such as borrower age, loan size, LTV, LTI, mortgage term, and neighborhood deprivation.

**Interpretation:** These regressions establish that borrowers with larger monetary gains are less likely to remain inactive, but the sensitivity is heterogeneous and incomplete.

### 4.2 Life-cycle model

The structural model is a life-cycle model of consumption, mortgage debt, house prices, income risk, and remortgaging. Each household faces decisions about consumption, savings, mortgage status, whether to act, whether to switch lender, whether to pass affordability checks, and whether to extract equity.

**Interpretation:** The model is necessary because a fixed pound cost means different things for different borrowers. A young low-income household with high marginal utility of consumption may require a much larger utility compensation for the same pound expenditure than an older high-income household.

### 4.3 Nonpecuniary costs

The model includes three nonpecuniary costs:

- $Z^{action}$: broad cost of taking action rather than remaining inactive;
- $Z^{switch}$: cost of switching lender;
- $Z^{checks}$: cost associated with affordability checks and documentation.

**Interpretation:** These costs are not direct product fees. They capture hassle, attention, time, paperwork, cognitive effort, and discomfort with financial engagement.

### 4.4 Model calibration

The model is calibrated to match moments by age group, including the fractions of borrowers who remain inactive, remortgage internally with or without equity extraction, and remortgage externally. It also targets equity extraction moments.

**Interpretation:** The calibration uses choice shares to infer action costs. If the model predicts too much remortgaging relative to the data, nonpecuniary costs must be higher.

### 4.5 Welfare calculations

The welfare calculations remove subsets of nonpecuniary costs and compute the compensating variation. Results are reported both in pounds and as a fraction of income.

**Interpretation:** Reporting welfare as a fraction of income is crucial. Pound costs may be larger for richer borrowers, but the burden relative to income is larger for young and lower-income households.

## 5 Identification and empirical design

### 5.1 What identifies action costs

The structural model identifies nonpecuniary costs by comparing observed remortgaging choices to the monetary gains available to each borrower. If large gains frequently fail to induce action, the model requires higher action costs to rationalize behavior.

**Interpretation:** The model is not identified from a randomized intervention. It is identified from the interaction between observed choice shares, measured monetary incentives, income heterogeneity, age heterogeneity, and the model's marginal utility structure.

### 5.2 Why the U.K. reset setting helps

The U.K. setting gives a clear decision date. At the end of the discounted period, the borrower faces a predictable shift to a higher reversion rate. This reduces ambiguity relative to standard refinancing studies, where households must forecast future interest rates and decide when to exercise an option.

**Interpretation:** This is the main empirical design advantage. Borrowers do not need sophisticated interest-rate models to know that staying inactive is costly.

### 5.3 Internal versus external remortgaging

The distinction between internal and external remortgages identifies different frictions. Internal remortgaging without equity extraction requires action but not lender switching or affordability checks. External remortgaging requires both action and lender switching. Internal remortgaging with equity extraction requires action and affordability checks.

**Interpretation:** Observing which remortgage option borrowers choose helps separate the broad cost of action from the more specific costs of switching and checks.

### 5.4 Threats to interpretation

- **Unobserved eligibility.** Some borrowers may be unable to remortgage for reasons not in the data. The sample restrictions reduce this risk.
- **Expectations.** Borrowers may expect to move, repay, or face future rate changes. The paper separates repayment and follows borrowers after the reset.
- **Measurement error in monetary gains.** Internal and external rates are estimated from comparable observed products. Tables 3 and 4 validate rate and fee inputs.
- **Model dependence.** Welfare costs depend on the life-cycle model and utility specification.
- **Policy external validity.** U.K. institutional details, such as short fixed-rate periods and internal product transfers, may differ from other mortgage markets.

**Interpretation:** The paper's strongest evidence is not a single regression coefficient. It is the joint consistency of measured gains, action patterns, remortgage timing, and structural welfare costs.

## 6 Tables and figures

### 6.1 Figure 1: Remortgage types and market practices; Figure 2: Timeline

**Read:** Figure 1 maps the choice set: inaction, internal remortgaging with/without equity extraction, and external remortgaging. Figure 2 shows the sample timing and which data source identifies each outcome.

**Economic message:** The figures establish the empirical design. They show why the authors can observe all relevant outcomes and why different choices map to different action costs.

**Interpretation:** Figure 1 is the friction map; Figure 2 is the measurement map. Together they explain why the U.K. setting is especially useful for estimating nonpecuniary action costs.

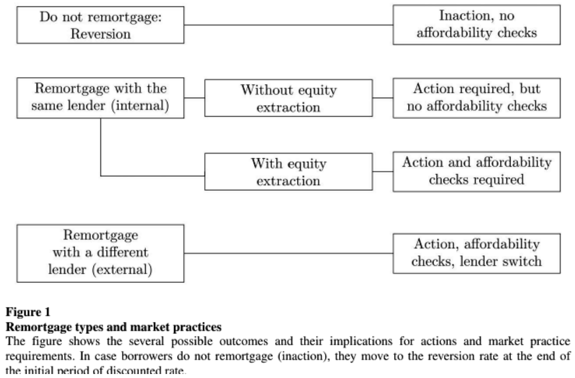



Figure 1: remortgage options and associated market-practice requirements.

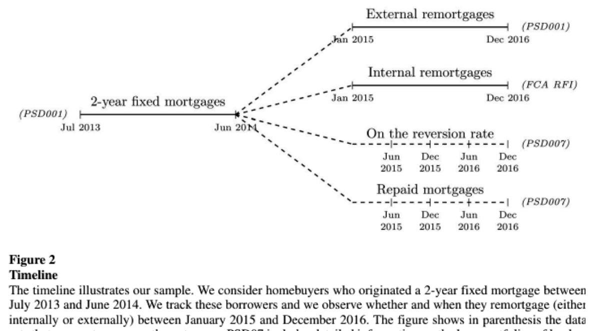



Figure 2: sample timeline and data sources used to measure outcomes.

The first pair of figures defines the decision tree and the measurement design.

## 6.2 Table 1: Remortgaging outcomes; Figure 3: Time of remortgage

**Read:** Table 1 shows that 19% remortgage externally, 47% internally, 20% remain on the reversion rate, and 14% repay before December 2016. Figure 3 shows timing: internal remortgages bunch around regular intervals and the reset date, while external remortgages have a longer right tail.

**Economic message:** A large share of borrowers take action, but a substantial minority remains inactive despite being eligible. Timing differences show that internal remortgaging is simpler and faster, while external remortgaging involves more delays.

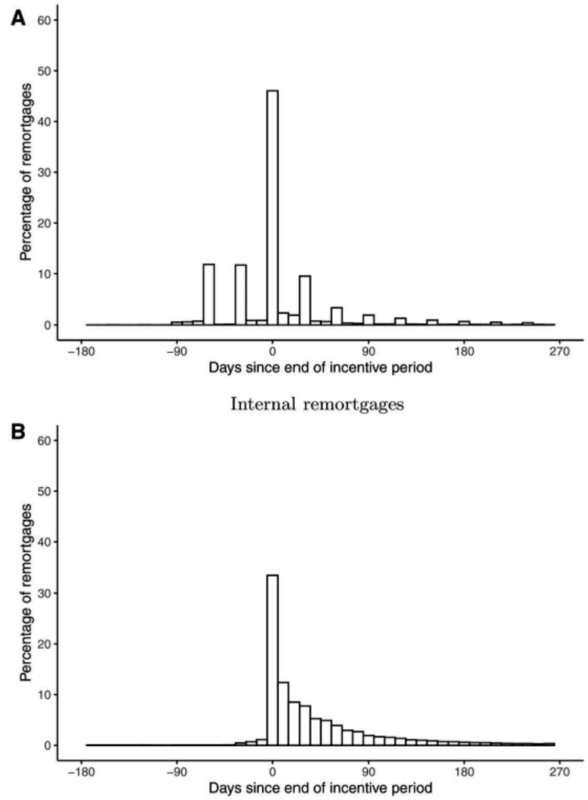
**Interpretation:** The outcome table gives the unconditional puzzle; the timing figure gives clues about frictions. External remortgages take longer because they require lender switching and documentation.

**Table 1**
**Remortgaging outcomes**

| Outcomes | *Number* | % |
|---|---|---|
| External remortgage | 52,677 | 19 |
| Internal remortgage | 125,836 | 47 |
| On reversion in Dec 2016 | 52,741 | 20 |
| Mortgage repaid before Dec 2016 | 37,436 | 14 |
| Total | 268,690 | 100 |

This table reports the remortgaging outcomes for those borrowers who took out a 2-year fixed rate mortgage for property acquisition between July 2013 and June 2014, grouping borrowers into four categories: (1) those who remortgaged with a different lender (external); (2) those who remortgaged with the same lender (internal); (3) those who did not remortgage and who were on the reversion rate in December 2016; and (4) those who did not remortgage but prepaid the loan by December 2016.

Table 1: outcome shares by December 2016.




External remortgages

**Figure 3**
**Time of remortgage**
The figure shows the distributions of the difference in number of days between the end of the discounted period and the date of the remortgage. A negative value means that borrowers remortgaged before the end of the discounted period. Panel A shows the results for internal remortgages, and panel B shows the results for external remortgages.

Figure 3: timing distributions for internal and external remortgages.

Outcome shares and timing jointly document the scale and type of inaction.

### 6.3 Table 2: Summary statistics at origination, by outcome

**Read:** Table 2 compares borrowers by eventual outcome. Inactive borrowers have lower income and live in more deprived neighborhoods. External remortgagors have larger loan values and somewhat higher leverage.

**Economic message:** The borrowers who remain on reversion rates are not randomly selected; they are more financially constrained and potentially more exposed to nonpecuniary action costs.

**Interpretation:** The table is important because the structural model must explain both levels of inaction and heterogeneity in inaction. Income and deprivation patterns motivate the later income-heterogeneity extension.

table2$_{o}rigination_{s}ummary.png$**Table 2: Summary statistics at origination, by outcome.**$Thistablecomparesborrowe$

## 6.4 Table 3: Estimated interest rate compared to realized rate; Table 4: Product fees

**Read:** Table 3 validates the rate estimation procedure for internal remortgages; median differences between estimated and realized rates are close to zero. Table 4 shows that external remortgage fees are much higher than internal remortgage fees.

**Economic message:** These tables validate key inputs into the gain calculations. If rates or fees were mismeasured, monetary gains and structural costs would be biased.

**Interpretation:** Table 3 supports the constructed counterfactual internal rate. Table 4 shows that fees are an important part of the external-versus-internal trade-off and help explain why external remortgages are less common.

**Table 3**
**The estimated interest rate compared to the realized rate, by borrower characteristics**

| Characteristics | 10th | 25th | Mean | Median | 75th | 90th |
|---|---|---|---|---|---|---|
| Unconditional | −0.25 | −0.10 | 0.02 | 0.00 | 0.10 | 0.35 |
| Borrower type | | | | | | |
| First time buyer | −0.20 | −0.05 | 0.04 | 0.00 | 0.15 | 0.40 |
| Home mover | −0.30 | −0.10 | −0.00 | 0.00 | 0.10 | 0.30 |
| Total gross income | | | | | | |
| Low | −0.10 | 0.00 | 0.08 | 0.00 | 0.20 | 0.40 |
| Medium | −0.20 | −0.05 | 0.04 | 0.00 | 0.15 | 0.35 |
| High | −0.40 | −0.15 | −0.04 | 0.00 | 0.05 | 0.25 |
| Loan value | | | | | | |
| Low | −0.10 | 0.00 | 0.07 | 0.00 | 0.20 | 0.40 |
| Medium | −0.15 | 0.00 | 0.06 | 0.00 | 0.15 | 0.40 |
| High | −0.40 | −0.15 | −0.06 | −0.00 | 0.05 | 0.24 |
| Joint income applicant | | | | | | |
| False | −0.25 | −0.05 | 0.03 | 0.00 | 0.15 | 0.40 |
| True | −0.25 | −0.10 | 0.02 | 0.00 | 0.10 | 0.30 |

This table compares the estimated internal remortgaging interest rate and the realized one. The former is the median rate by: (a) quarter of the expiry of the initial period of discounted rates; (b) revised LTV (calculated with the revised house price at expiry date); and (c) lender. The latter is taken directly from our data, from those borrowers who did remortgage internally using the same loan type and without equity extraction. The first panel shows several percentiles of the difference for the whole sample, and the remaining panels when we condition on origination characteristics. High, medium, and low are defined according to the terciles of the corresponding distributions.

Table 3: validation of estimated internal rates.

**Table 4**
**Summary statistics on product fees of remortgaged loans**

| | Observations | *Mean* | SD | p10 | Median | p90 |
|---|---|---|---|---|---|---|
| External remortgages | 53,449 | 528.56 | 567 | 0 | 370 | 1,089 |
| Internal remortgages | 121,797 | 79.09 | 217 | 0 | 0 | 225 |
| *A. Unconditional fees (in pounds)* | | | | | | |
| | Observations | *Mean* | SD | p10 | Median | p90 |
| Fees for loans with amount smaller than the median | | | | | | |
| External remortgages | 20,679 | 208.65 | 370 | 0 | 0 | 999 |
| Internal remortgages | 66,585 | 43.25 | 121 | 0 | 0 | 195 |
| Fees for loans with amount larger than the median | | | | | | |
| External remortgages | 32,770 | 730.42 | 577 | 0 | 995 | 1,260 |
| Internal remortgages | 55,202 | 122.32 | 289 | 0 | 0 | 225 |
| *B. Fees conditional on loan amount* | | | | | | |
| | Observations | *Mean* | SD | p10 | Median | p90 |
| Fees for loans with interest rate lower than the median | | | | | | |
| External remortgages | 34,194 | 639.56 | 581 | 0 | 810 | 1,225 |
| Internal remortgages | 53,415 | 110.28 | 270 | 0 | 0 | 225 |
| Fees for loans with interest rate higher than the median | | | | | | |
| External remortgages | 19,255 | 331.42 | 482 | 0 | 30 | 1,025 |
| Internal remortgages | 68,329 | 54.75 | 161 | 0 | 0 | 195 |
| *C. Fees conditional on loan interest rate* | | | | | | |

The table shows mean, standard deviation and selected percentiles of product fees (in pounds) for internally and externally remortgaged loans in our data. The first panel shows the results for all remortgages of a given type, and the remaining panels conditional on the amount and interest rate of the remortgaged loan.

Table 4: product-fee summary statistics.

Rate and fee validation underpins the monetary-gain measures.

## 6.5 Table 5: Monetary gains of remortgaging net of fees; Table 6: Inaction regressions

**Read:** Table 5 reports the net monetary gains from internal and external remortgaging. Table 6 regresses inaction on gains, income, and controls. Borrowers are less likely to remain inactive when gains are high, but inaction remains substantial.

**Economic message:** Borrowers respond to monetary incentives, but not enough to eliminate inaction. This is the reduced-form motivation for nonpecuniary costs.

**Interpretation:** The key reading rule is to compare gains to behavior. If gains are large and inaction persists, a rational model needs a large action cost, a high marginal utility of cash, or both.

**Table 5**
**Annual monetary gains of remortgaging internally and externally**

| | Observations | Mean | p10 | Median | p90 |
|---|---|---|---|---|---|
| *A. Monetary gains based on internal remortgages* | | | | | |
| Annual gains (£) | | | | | |
| All | 221,314 | 3,048 | 1,000 | 2,293 | 5,768 |
| Internal | 121,726 | 2,974 | 1,054 | 2,324 | 5,518 |
| External | 48,244 | 3,985 | 1,248 | 2,864 | 7,825 |
| Reversion | 51,344 | 2,343 | 800 | 1,788 | 4,386 |
| Annual gains (% of income) | | | | | |
| All | 221,314 | 6.05 | 3.03 | 5.60 | 9.15 |
| Internal | 121,726 | 6.06 | 3.15 | 5.70 | 9.07 |
| External | 48,244 | 6.83 | 3.41 | 5.99 | 10.41 |
| Reversion | 51,344 | 5.29 | 2.55 | 5.02 | 8.36 |
| *B. Monetary gains based on external remortgages* | | | | | |
| Annual gains (£) | | | | | |
| All | 230,991 | 3,385 | 1,179 | 2,620 | 6,364 |
| Internal | 125,674 | 3,308 | 1,240 | 2,620 | 6,073 |
| External | 52,672 | 4,319 | 1,565 | 3,386 | 8,131 |
| Reversion | 52,645 | 2,635 | 918 | 2,027 | 4,914 |
| Annual gains (% of income) | | | | | |
| All | 230,991 | 6.73 | 3.57 | 6.53 | 9.86 |
| Internal | 125,674 | 6.73 | 3.67 | 6.56 | 9.72 |
| External | 52,672 | 7.51 | 4.21 | 7.20 | 11.08 |
| Reversion | 52,645 | 5.95 | 2.88 | 5.78 | 9.01 |

The table shows the annual gains from remortgaging internally (panel A) and externally (panel B). We define gains as the difference between the reversion rate and the estimated rate, times the balance at the end of the discounted period, minus one half of the one-off remortgaging fees (so that they are spread over the 2-year period of discounted rates). The estimated rate is the interest rate that the borrower would have been able to obtain for a remortgage with the same lender (internal) or with any lender (external). The table shows the annual gains in pounds and as a percentage of the borrower's origination gross income. In each panel, the first row shows the gains for all borrowers, and the remaining rows show the gains conditional on the outcome (remortgage internal and external, and reversion).

to external remortgagors in the same geographical region. We assume that an individual would be able to get a deal with the median interest rate and the corresponding fees.

Panel B of Table 5 shows that the estimated monetary gains of external remortgaging are on average higher than those of internal remortgaging. For instance, among all borrowers the mean values are £3,385 and £3,048, respectively. Furthermore, the estimated gains of remortgaging externally are higher for those borrowers who ended up doing so (equal to £4,319) than for those borrowers who remortgaged internally (£2,974, shown in the second row of panel A). It is important to note that since an external remortgage is always treated as a new loan, it requires an affordability assessment and a full property valuation. Therefore, unlike for internal remortgages, we do not know whether borrowers would have been able to remortgage externally.

**2.3 Regression analysis for inaction**

We analyze the drivers of inaction using reduced form regression analysis. The dependent variable is a dummy equal to 1 if a consumer ends the sample period on reversion. We exclude consumers who repay their mortgage, so that the dummy equals zero if a consumer remortgages, either internally or externally.

Table 5: net-of-fees monetary gains from remortgaging.

**Table 6**
**Regression for the likelihood of inaction**

| | Dependent variable: *Inaction* | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| First-time buyer | 0.008*** | 0.009*** | 0.007*** | 0.001 |
| | (0.002) | (0.002) | (0.002) | (0.002) |
| Joint income | − 0.024*** | − 0.025*** | − 0.024*** | − 0.016*** |
| | (0.002) | (0.002) | (0.002) | (0.002) |
| Age in [30, 40) | − 0.004** | − 0.005** | − 0.005** | − 0.006*** |
| | (0.002) | (0.002) | (0.002) | (0.002) |
| Age in [40, 50) | 0.015*** | 0.014*** | 0.014*** | 0.011*** |
| | (0.003) | (0.003) | (0.003) | (0.003) |
| Age 50 or higher | 0.057*** | 0.055*** | 0.054*** | 0.050*** |
| | (0.004) | (0.004) | (0.004) | (0.004) |
| Income and gains grid | Yes | Yes | Yes | Yes |
| Year-month origination FE | No | Yes | Yes | Yes |
| Postcode area FE | No | No | Yes | Yes |
| Lender FE | No | No | No | Yes |
| Observations | 221,314 | 221,314 | 221,314 | 221,314 |
| $R^2$ | .046 | .049 | .050 | .072 |
| Adjusted $R^2$ | .046 | .049 | .049 | .071 |

The dependent variable is a dummy variable that takes a value of one if the borrower does not remortgage (inaction), and zero if the borrower remortgages either internally or externally by December 2016. Borrowers who repaid the loan are not included. Throughout we include as explanatory variables, 100 dummy variables that take the value of one if the borrower is in a given cell of the $10 \times 10$ matrix of monetary gains of remortgaging and of income shown in the previous table, and zero otherwise. The columns differ in the borrower characteristics and in the sets of fixed effects included. The model is estimated by ordinary least squares. $^{*}p<.1$; $^{**}p<.05$; $^{***}p<.01$.

In our sample, all borrowers would have been able to remortgage internally. In the first column of Table 6, we regress the inaction dummy on 100 dummies corresponding to the 100 cells of the $10 \times 10$ matrix of the internal remortgaging gains (in £) and income (i.e., each explanatory variable takes the value of one for borrowers in that cell, and zero otherwise). We also include as explanatory variables a dummy for first-time buyer (FTB vs. home-mover), a dummy for single (vs. joint applicant) and 3 age dummies (the base group is those younger than 30 years old).

We see that, controlling for remortgaging gains and income, FTBs are more

Table 6: inaction regressions by gains and characteristics.

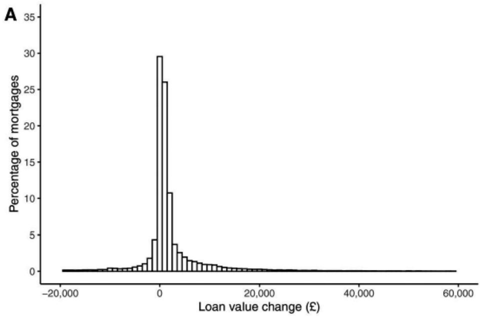
The reduced-form tables show that gains matter but do not fully explain action.

## 6.6 Figure 4: Equity extraction for internal and external remortgagors; Figure 5: House price increases and equity extraction

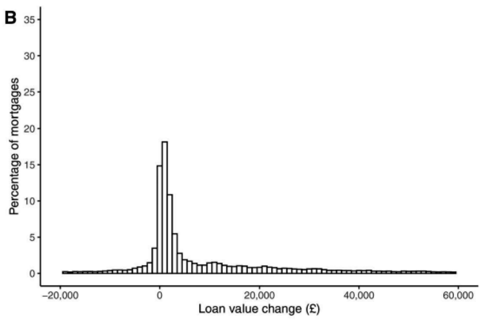
**Read:** Figure 4 shows that equity extraction is concentrated near zero but has long right tails, especially for external remortgagors. Figure 5 relates equity extraction to house price appreciation and shows that rising house prices facilitate extraction.

**Economic message:** Equity extraction is a motive for taking action, especially externally. House-price gains relax collateral constraints and make extraction more attractive.

**Interpretation:** These figures motivate why the structural model includes house prices, mortgage balances, and equity-extraction decisions. Borrowers are not only minimizing interest rates; some also rebalance mortgage debt and housing equity.



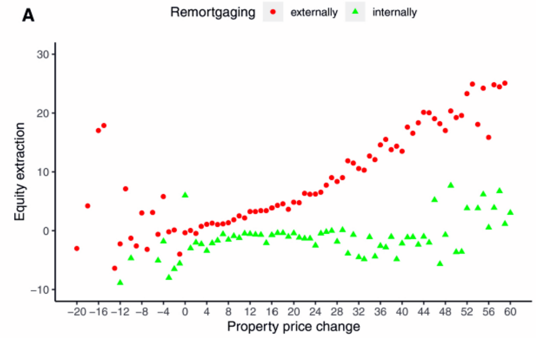

Internal remortgagors



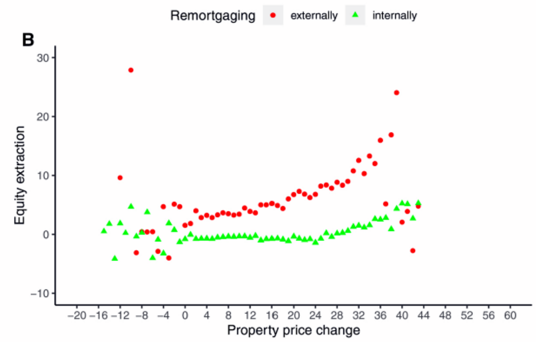

External remortgagors

**Figure 4**
**Equity extraction for internal and external remortgagors**
The figure plots the percentage of remortgaged loans with a given amount of equity extracted. Loan value change (or equity extracted) is calculated as the difference between the loan amount recorded in PSD1/RFI for the remortgage minus the remaining balance at the end of the discounted period of the previous mortgage. Positive values indicate equity extractions.

Figure 4: distribution of loan-balance changes.




Equity extraction vs. house price growth (PSD001/RFI)




Equity extraction vs. house price growth (HMLR)

**Figure 5**
**Equity extraction and property price growth**
On the vertical axis, equity extraction is measured as the log difference between the amount of the remortgaged loan and the outstanding loan amount at the end of the discounted period (i.e., before the remortgaging) multiplied by 100. On the horizontal axis, property price change is measured as the log difference between the property price at end of the discounted period and the origination price multiplied by 100. Figure 5, panel A, uses property

Figure 5: house-price increases and equity extraction.

Equity extraction connects remortgaging to household balance-sheet management.

## 6.7 Table 7: Model parameters; Table 8: Calibrated moments and targets

**Read:** Table 7 reports parameters for income processes, house prices, lender offer dispersion, and other primitives. Table 8 reports data moments targeted in calibration, including age-specific remortgage outcomes and equity extraction.

**Economic message:** The structural model is disciplined by both exogenous processes and observed mortgage-choice moments. The calibration must match not just inaction, but the whole distribution of remortgage actions.

**Interpretation:** These tables are the bridge from descriptive evidence to welfare. The model's inferred nonpecuniary costs are meaningful only if the model can reproduce the main data moments.

**Table 7**
**First-stage estimation and other model parameters**

| | | |
|---|---|---|
| *A. Income* | | |
| Standard dev. permanent shocks | $\sigma_\eta$ | 0.10 |
| Standard dev. temporary shocks | $\sigma_\epsilon$ | 0.15 |
| Autoregression coefficient | $\phi_v$ | 0.82 |
| *B. House prices* | | |
| Mean log growth | $\mu_H$ | 0.02 |
| Standard deviation | $\sigma_\zeta$ | 0.08 |
| *C. Interest rates* | | |
| Mean log interest rate | $\mu_r$ | 0.009 |
| Standard deviation | $\sigma_r$ | 0.025 |
| Autoregression coefficient | $\phi_r$ | 0.830 |
| Loan premium, internal | $\theta^{\text{internal}}$ | 0.023 |
| Loan premium, external | $\theta^{\text{external}}$ | 0.021 |
| Loan premium, reversion | $\theta^{\text{reversion}}$ | 0.040 |
| *D. Other model parameters* | | |
| Lock-in period | $lock$ - $in$ | 2 years |
| Mortgage maturity | $T^m$ | 30 years |
| Maximum loan-to-value | $LTV^{\max}$ | 0.90 |
| Maximum loan-to-income | $LTI^{\max}$ | 4.5 |
| Tax rate | $\tau$ | 0.20 |
| Minimum consumption | $\underline{C}$ | £ 3,000 |
| Maintenance expenses | $\alpha$ | 0.02 |
| Risk aversion | $\gamma$ | 2 |

The table reports model parameter values from the first-stage estimation and other model parameters. The income data are from the British Household Panel Survey. The house price data are from the Office of National Statistics. The interest rate data are from the Bank of England and the mortgage premium are from the mortgage data. The maximum LTV and LTI are based on market data and regulations.

consumer price index and calculate the mean and standard deviation of the logarithm of real house price changes reported in panel B of Table 7.

We use Bank of England data to parameterize the interest rate process. More precisely, we use annual data on the 3-month Treasury bill that we deflate using

Table 7: model parameters.

**Table 8**
**Target data and model moments**

| | Age groups 30–39 | 40–49 | 50–59 |
|---|---|---|---|
| *A. Data moments* | | | |
| Fraction reversion | 0.20 | 0.24 | 0.31 |
| Fraction internal without eq. extraction | 0.45 | 0.44 | 0.42 |
| Fraction internal with eq. extraction | 0.09 | 0.13 | 0.16 |
| Fraction external | 0.25 | 0.19 | 0.12 |
| Equity extraction (£1k) | 8.4 | 8.3 | 7.9 |
| *B. Model moments* | | | |
| Fraction reversion | 0.21 | 0.24 | 0.31 |
| Fraction internal without eq. extraction | 0.44 | 0.42 | 0.47 |
| Fraction internal with eq. extraction | 0.12 | 0.12 | 0.11 |
| Fraction external | 0.24 | 0.22 | 0.12 |
| Equity extraction (£1k) | 4.3 | 2.7 | 6.3 |

We divide borrowers into groups based on their age at origination. Panel A reports the data moments that we target in the model calibration: the fraction of inactive borrowers and of each remortgaging type and the average equity extraction. Panel B reports the corresponding model moments calculated from the simulated data.

cohort reaches the terminal age and drops out of the sample. This means that, in each period, we have in our simulated economy agents of different ages, with different income, houses, mortgages, financial savings and so on.

The mortgage data target moments reflect the choices of different borrowers at the end of the initial period of discounted rates, having purchased a house 2 years before those choices. We use the model simulated data that corresponds to agents that match those which we observe in the actual data, that is, those that in the simulated data bought a house 2 years before and are now making remortgaging decisions. In the actual data, the period between the house purchase and the remortgaging decisions was characterized by low interest rates and generally increasing house prices. Therefore, in simulated data, for these same periods, we set the realization of interest rates to low and the realizations of house prices to match the average observed in the data. We give

Table 8: data moments and calibrated model fit.

Model primitives and targeted moments used for structural calibration.

## 6.8 Figure 6: Data and model moments; Table 9: Equity extraction

**Read:** Figure 6 compares model-generated moments with the data by age group. Table 9 compares equity extraction in the data and model.

**Economic message:** The model captures the broad age patterns in inactivity, internal remortgaging, external remortgaging, and equity extraction. This supports its use for counterfactual welfare analysis.

**Interpretation:** The figure and table are validation checks. The model does not need to match every cell perfectly, but it must get the main life-cycle and equity-extraction patterns right.

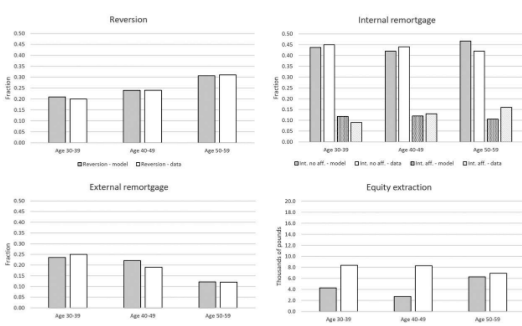



**Figure 6**
**Model moments and data moments for the different age groups**
The figure compares the data moments to those generated by the model for the calibrated parameters, for each of the three age groups. The figure plots the proportions of borrowers who remain inactive (reversion), who remortgage internally with and without equity extraction, and who remortgage externally. The last panel plots the average amount of equity extracted (in thousands of pounds).

extraction relative to the same lender with equity extraction. And $Z_{it}^{\text{switch}}$ affects primarily the proportion of agents who remortgage externally

Figure 6: model fit for targeted moments.

**Table 9**
**Nontargeted moments**

| | Age groups 30–39 | 40–49 | 50–59 |
|---|---|---|---|
| *A.Data moments* | | | |
| Wealth accumulation (£1k) | 72.6 | 165.0 | 246.5 |
| Equity extracted by internal rem. (£1k) | 2.86 | 2.85 | 3.82 |
| Equity extracted by external rem. (£1k) | 23.9 | 22.9 | 16.9 |
| House price change if no eq. extraction | 0.085 | 0.079 | 0.076 |
| House price change if eq. extraction | 0.094 | 0.085 | 0.082 |
| *B. Model moments* | | | |
| Wealth accumulation (£1k) | 81.7 | 141.6 | 270.2 |
| Equity extracted by internal rem. (£1k) | 3.14 | 2.50 | 7.70 |
| Equity extracted by external rem. (£1k) | 10.79 | 6.18 | 15.35 |
| House price change if no eq. extraction | 0.074 | 0.075 | 0.072 |
| House price change if eq. extraction | 0.077 | 0.077 | 0.063 |
| *C. Calibrated model parameters* | | | |
| Cost of action $Z_{it}^{\text{action}}$ | 0.00250 | 0.00112 | 0.00140 |
| Cost of checks $Z_{it}^{\text{checks}}$ | 0.00033 | 0.00000002 | 0.000025 |
| Cost of switch $Z_{it}^{\text{switch}}$ | 0.00027 | 0.00023 | 0.00029 |
| Preference parameters $\beta_i$ =0.92 and $\phi_i$ = 100 | | | |

The table compares nontargeted data (panel A) and model generated (panel B) moments. The moments are: (a) wealth accumulation; (b) the mean pound value of equity extracted by those remortgaging internally and externally; and (c) the (annual) average house price change experienced by borrowers who do not extract and who extract home equity. Panel C reports the values for the calibrated parameters.

Table 9: equity extraction moments in data and model.

Calibration fit for remortgaging outcomes and equity extraction.

## 6.9 Table 10: Welfare gains; Figure 7: Policy functions

**Read:** Table 10 reports welfare gains from eliminating nonpecuniary costs. Figure 7 plots representative policy functions for remortgaging choices as a function of cash on hand.

**Economic message:** The largest welfare gains come from removing the broad cost of action. Policy functions show that remortgage choice depends on liquidity, income, and housing wealth.

**Interpretation:** Table 10 gives the paper's main welfare numbers. Young borrowers experience particularly large gains as a fraction of income. Figure 7 provides intuition: action costs shift the thresholds at which households refinance or remain inactive.

**Table 10**
**Welfare gains**

| Nonpecuniary costs eliminated | Age groups 30–39 | 40–49 | 50–59 |
|---|---|---|---|
| *A. In pounds* (£1k) | | | |
| Switch ($Z_{it}^{switch}$=0 ) | 0.441 | 0.575 | 0.347 |
| Switch and checks ($Z_{it}^{switch}$,$Z_{it}^{checks}$=0 ) | 0.626 | 0.619 | 0.385 |
| Action, switch, and checks ($Z_{it}^{action}$,$Z_{it}^{switch}$,$Z_{it}^{checks}$=0 ) | 4.793 | 4.951 | 3.302 |
| *B. Fraction of income* | | | |
| Switch ($Z_{it}^{switch}$=0 ) | 0.018 | 0.018 | 0.008 |
| Switch and checks ($Z_{it}^{switch}$,$Z_{it}^{checks}$=0 ) | 0.026 | 0.019 | 0.009 |
| Action, switch, and checks ($Z_{it}^{action}$,$Z_{it}^{switch}$,$Z_{it}^{checks}$=0 ) | 0.191 | 0.149 | 0.079 |

The table reports welfare measures of the importance of the nonpecuniary remortgaging costs. We solve the model for the base case and for an alternative scenario in which some of the nonpecuniary costs are set to zero, while the others are kept at the base value. We calculate the cash that individuals in the economy where the costs are positive must be given so that they are as well off in utility terms as in the economy with zero costs. We perform the calculations for each individual in our simulated data, and calculate averages by age. Panel A shows the results in pounds (panel B as a proportion of income). In each panel, the first row reports the welfare gains of setting nonpecuniary costs of switching to zero, the second row of setting the costs of switching and of checks to zero, and the last row of setting all the costs to zero.

in Table 8 we have previously shown that older borrowers are more likely to remain inactive. Older individuals tend to have lower monetary gains from

Table 10: welfare gains from removing nonpecuniary costs.

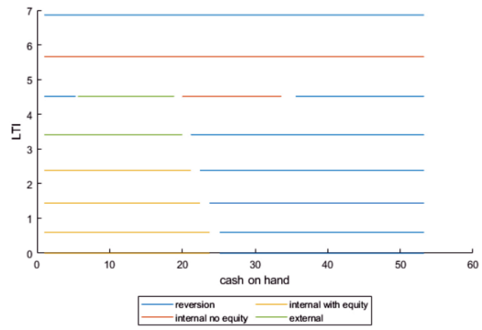



**Figure 7**
**Model policy functions**
The figure plots representative policy functions for remortgaging outcomes in our model as a function of cash-on-hand (horizontal axis) and loan outstanding relative to permanent income (vertical axis).

that inaction is for agents utility wise, as measured by the welfare gains of eliminating the nonpecuniary costs of action.

Figure 7: representative policy functions.

Welfare and policy functions explain the structural meaning of action costs.

## 6.10 Figure 8: Out-of-sample test; Table 11 and Figure 9: Model extensions

**Read:** Figure 8 compares data and model moments for a later out-of-sample cohort. Table 11 reports welfare gains in model extensions with an inaction shock and income-dependent action costs. Figure 9 shows that the income-heterogeneous model matches different outcomes for low- and high-income borrowers.

**Economic message:** The out-of-sample test and extensions show that the baseline mechanism travels beyond the main calibration sample, while income heterogeneity improves the fit for low-income borrowers.

**Interpretation:** Table 11 is important because it separates pure inattention shocks from nonpecuniary costs. Even with a 10% inaction shock, significant nonpecuniary costs are still needed. In the income-extension model, costs are larger as a fraction of income for low-income households.

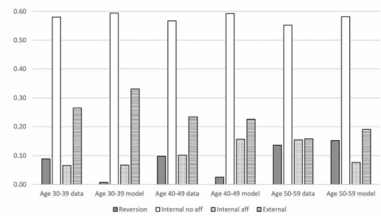



**Figure 8**
**Model and data moments for the out-of-sample test**
The figure compares data moments to those generated by the model for the out-of-sample test. The figure plots the proportions of borrowers who remain inactive (reversion), who remortgage internally with and without equity extraction, and who remortgage externally.

In terms of the drivers of the changes, the lower rate of house price appreciation lead to inaction, but the higher reversion rate and in initial leverage incentivize action and dominate in this out-of-sample check. The lower rate of house price appreciation also reduces the likelihood of equity extraction outcomes.

Figure 8: out-of-sample model fit.

**Table 11**
**Model extensions, welfare gains**

| Nonpecuniary costs eliminated | Age groups 30–39 | 40–49 | 50–59 |
|---|---|---|---|
| *A. Model with inaction shock ($\pi^{inaction}$=0.10)* | | | |
| *A.1: In pounds (£1k)* | | | |
| Switch ($Z_{it}^{switch}$=0 ) | 0.433 | 0.576 | 0.346 |
| Switch and checks ($Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 0.648 | 0.587 | 0.343 |
| Action, switch, and checks ($Z_{it}^{action}$, $Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 4.028 | 4.514 | 2.958 |
| *A.2: Fraction of income* | | | |
| Switch ($Z_{it}^{switch}$=0 ) | 0.017 | 0.018 | 0.008 |
| Switch and checks ($Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 0.026 | 0.018 | 0.008 |
| Action, switch, and checks ($Z_{it}^{action}$, $Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 0.159 | 0.135 | 0.071 |
| *B. Model with two income groups* | | | |
| *B.1.1: Low-income group, in pounds (£1k)* | | | |
| Switch ($Z_{it}^{switch}$=0 ) | 0.236 | 0.414 | 0.272 |
| Switch and checks ($Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 0.436 | 0.425 | 0.269 |
| Action, switch, and checks ($Z_{it}^{action}$, $Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 3.394 | 3.297 | 2.267 |
| *B.1.2: Low-income group, fraction of income* | | | |
| Switch ($Z_{it}^{switch}$=0 ) | 0.014 | 0.019 | 0.010 |
| Switch and checks ($Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 0.026 | 0.019 | 0.009 |
| Action, switch, and checks ($Z_{it}^{action}$, $Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 0.200 | 0.145 | 0.079 |
| *B.2.1: High-income group, in pounds (£1k)* | | | |
| Switch ($Z_{it}^{switch}$=0 ) | 0.690 | 0.806 | 0.541 |
| Switch and checks ($Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 1.105 | 0.914 | 0.541 |
| Action, switch, and checks ($Z_{it}^{action}$, $Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 6.834 | 6.225 | 3.903 |
| *B.2.2: High-income group, fraction of income* | | | |
| Switch ($Z_{it}^{switch}$=0 ) | 0.019 | 0.017 | 0.009 |
| Switch and checks ($Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 0.030 | 0.019 | 0.009 |
| Action, switch, and checks ($Z_{it}^{action}$, $Z_{it}^{switch}$, $Z_{it}^{checks}$=0 ) | 0.183 | 0.126 | 0.062 |

The table reports welfare measures of the importance of the nonpecuniary remortgaging costs for the model extensions with an exogenous inaction shock (panel A) and with nonpecuniary remortgaging costs that depend on both age and income (panel B). In the former, in each period agents face an i.i.d. probability that they are hit by a shock such that they remain inactive regardless of what the optimal decision would be in case they were not hit by the shock. For each scenario, we solve the model for the base case and for an alternative scenario in which

Table 11: welfare gains in model extensions.

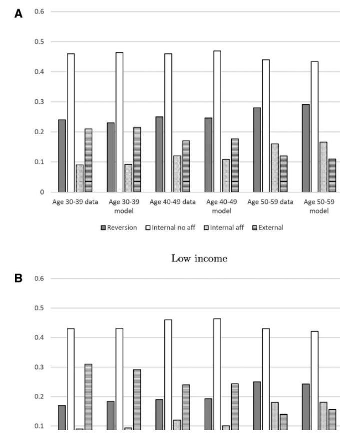



Figure 9: income-heterogeneous model fit.

Out-of-sample and extension evidence for the robustness of inferred action costs.

# 7 Short summary

- The paper studies U.K. borrowers whose 2-year discounted mortgage rate ends.
- Borrowers can remortgage internally, remortgage externally, stay inactive on the reversion rate, or repay.
- The data combine PSD001, an FCA RFI on internal remortgages, and PSD007 loan-book information.
- About one in five eligible borrowers remains on the reversion rate by December 2016.
- Monetary gains from remortgaging are often large, even net of fees.
- Borrowers respond to monetary gains, but inaction remains widespread.
- A life-cycle model infers significant nonpecuniary costs of action.
- These costs include a broad action cost, lender-switching cost, and affordability-check cost.
- Welfare gains from eliminating action costs are large, especially for young and lower-income households.
- Out-of-sample and income-heterogeneity extensions support the main mechanism.

# 8 Conclusion

## 8.1 Core conclusion

The paper concludes that household inaction in the mortgage market reflects substantial nonpecuniary costs of action. Even in a setting where monetary gains from remortgaging are salient and relatively easy to compute, many borrowers fail to act. A structural model implies that these costs are economically meaningful and heterogeneous across the life cycle and income distribution.

## 8.2 Economic intuition

The intuition is that remortgaging is not a single financial calculation. It is a bundle of tasks: noticing the rate reset, comparing products, contacting a lender, providing paperwork, possibly switching lender, and deciding whether to extract equity. For households with high marginal utility of consumption or low financial confidence, these nonmonetary burdens can offset large pound savings.

## 8.3 Policy implications

Policies that reduce action costs can have high welfare returns. Examples include clearer communications before rate resets, simplified internal product transfers, automatic prompts, standardized comparison tools, or targeted help for low-income borrowers. The structural results suggest that reducing broad action costs matters more than reducing only lender-switching or affordability-check costs.

## 8.4 Incremental contribution revisited

The paper's distinctive contribution is to measure action costs using a setting where the value of action is unusually transparent. This makes the finding more striking: if households are inactive even when payment savings are clear, nonpecuniary frictions are likely important in many other consumer-finance decisions as well.

## 8.5 Limitations and extensions

The estimates depend on the life-cycle model and the assumed utility structure. The setting is specific to the U.K. mortgage market, where short discounted periods and internal product transfers are common. The model captures major features of remortgaging but cannot observe all expectations, search behavior, communications, and psychological frictions. Future work could study lender-specific reminders, randomized disclosures, or direct measures of attention and financial confidence.

## 8.6 Takeaway

The main takeaway is that financial inaction can be rationalized only by recognizing large nonpecuniary action costs. Consumer-finance policy should therefore focus not only on prices and fees, but also on the practical and cognitive burden of taking action.